POLITECNICO
MILANO 1863

Pneumonia Detection Using Anchor-Free vs. Anchor-Based Object Detection

Reproducing and Extending Wu et al. (2024)

Numerical Analysis for Machine Learning — Politecnico di Milano

Mohamed Z. M. Mandour
mohamedzeyad.mandour@mail.polimi.it

Elena Nuttini
elena.nuttini@mail.polimi.it

June 2026

Abstract

This report reproduces and extends the method proposed by Wu et al. [1], which applies the anchor-free FCOS (Fully Convolutional One-Stage) detector to pneumonia detection on chest X-rays from the RSNA Pneumonia Detection Challenge dataset. We implement and compare four detector variants sharing the same ResNet-50 + Feature Pyramid Network (FPN) backbone: (1) FCOS with the paper's anchor-free head trained under our modern Adam recipe, (2) the same FCOS head retrained under the paper's original SGD+momentum recipe as a controlled ablation, (3) RetinaNet as a one-stage anchor-based baseline, and (4) Faster R-CNN as a two-stage anchor-based baseline. Training is performed on the full RSNA dataset (26,684 images) for 40 epochs on four NVIDIA H100 80 GB SXM GPUs (one model per GPU) with BF16 mixed-precision, cosine annealing, warm-up, and Exponential Moving Average (EMA) of the weights. At evaluation time we apply horizontal-flip Test-Time Augmentation (TTA) and Gaussian Soft-NMS, and combine FCOS with RetinaNet via Weighted Box Fusion (WBF).

All detectors *exceed* the paper's reported FCOS AP@0.5 of 28.5%: our FCOS (Adam) reaches 35.3% (+6.8), FCOS with the paper's SGD recipe reaches 39.8% (+11.3), RetinaNet reaches 41.2% (+12.7), and Faster R-CNN reaches 42.9% (+14.4). At the patient level, the WBF ensemble of FCOS and RetinaNet delivers the best classification F1 of 63.0% at the fixed confidence threshold of 0.3, while Faster R-CNN reaches a single-model F1 of 53.2% at the same threshold. Our central empirical finding is that, under a shared backbone and matched training recipe, the anchor-based detectors *outperform* the anchor-free FCOS under Adam, but the gap collapses when FCOS is trained with the paper's original SGD recipe: FCOS (paper SGD) at 39.8% has a 95% patient-bootstrap CI of $[37.1, 43.1]$ that overlaps RetinaNet's $[38.2, 44.1]$, so under matched training budget the two are statistically indistinguishable on this split. Faster R-CNN remains marginally on top (+1.69 pts vs. RetinaNet, paired-bootstrap $p = 0.080$). We argue that this refines rather than invalidates Wu et al.'s paradigm argument: once modern training recipes (mixed precision, cosine schedule, EMA, medical-specific augmentation) are applied uniformly to all baselines, the choice of detection head is no longer the dominant accuracy lever, and the anchor-based detectors converge faster on a fixed epoch budget. To make the single-run conclusion defensible without further retraining, Section 10 adds seven post-hoc analyses on the cached per-image predictions — patient-level bootstrap CIs on AP@0.5, a three-way split protocol that fits the patient-level threshold on a held-out calibration half, RSNA-percentile size buckets, FROC with the standard CPM, reliability diagrams with ECE, a 5-fold-CV learnt patient-level aggregator that replaces the naive "any box $> \tau$" rule, and a paired bootstrap of Faster R-CNN vs. RetinaNet.

1 Introduction

Pneumonia is one of the leading causes of death worldwide, particularly among children and the elderly. Accurate and timely detection from chest radiographs is critical but challenging, as pneumonia opacities can be subtle, overlapping with other pathologies, and difficult to localize precisely.

Deep learning–based object detection has shown great promise for automating pneumonia detection. Most prior approaches use *anchor-based* detectors such as Faster R-CNN [6] or RetinaNet [3], which rely on predefined anchor boxes to propose candidate regions. In contrast, Wu et al. [1] proposed using FCOS [2], an *anchor-free* detector that directly predicts bounding boxes at each spatial location of the feature map, eliminating the need for hand-designed anchor configurations.

The key advantages of the anchor-free approach include:

- No hyperparameter tuning for anchor shapes, sizes, or aspect ratios.
- More flexible localization, particularly for irregularly-shaped or variable-size lesions.
- Simpler architecture with fewer design choices.

In this project, we implement and compare three detection architectures:

1. **FCOS** — the paper's proposed anchor-free detector,
2. **RetinaNet** — a one-stage, anchor-based detector with focal loss,
3. **Faster R-CNN** — a two-stage, anchor-based detector with RPN.

All models share a ResNet-50 backbone with Feature Pyramid Network (FPN), ensuring that performance differences arise from the detection paradigm rather than feature extraction capacity.

Contributions

- A faithful reproduction of Wu et al. [1] on the full RSNA dataset: our FCOS-Adam crosses the paper's reported 28.5% AP@0.5 at epoch 12 and ends +6.8 points above it at the 40-epoch convergence; the controlled SGD-recipe ablation reaches +11.3 points above the paper number.
- A fair, same-backbone, same-budget comparison of three representative detection paradigms (anchor-free FCOS, one-stage anchor-based RetinaNet, two-stage anchor-based Faster R-CNN), with FCOS evaluated under both our Adam recipe and the paper's original SGD recipe. The Adam-trained FCOS is decisively below the anchor-based detectors on AP@0.5; under the paper SGD recipe the FCOS-vs.-RetinaNet gap collapses into the patient-bootstrap noise.
- A Weighted Box Fusion ensemble of the two one-stage detectors (FCOS + RetinaNet) that improves patient-level F1 by a substantial margin without retraining, while being cheaper than the two-stage Faster R-CNN baseline.
- A detailed analysis of the score-calibration mismatch between the four detectors that explains why fixed-threshold patient-level F1 differs much more than threshold-free ROC AUC — a confound that the paper does not address — together with a learnt 5-fold cross-validated aggregator that essentially closes the F1 gap.
- A public PyTorch implementation with reproducible training recipes (cosine + warm-up schedule, EMA, medical-imaging augmentation pipeline) and all evaluation artefacts released with the report.

Outline. Section 2 surveys object detection and its use in medical imaging. Section 3 introduces the mathematical background: FPN, focal loss, EMA, (Soft-)NMS, and WBF. Section 4 describes the RSNA dataset and our preprocessing. Section 5 presents the three detector heads formally and Section 6 details the training recipe. Section 7 defines the metrics. Section 8 reports the quantitative and qualitative results, Section 9 provides an analysis of each component's contribution, Section 11 discusses the findings including the perhaps surprising anchor-based win, and Section 12 covers limitations and future directions.

2 Related Work

Two-stage anchor-based detection. The modern two-stage detector lineage begins with R-CNN [8], which extracts region proposals with Selective Search and classifies each proposal with a CNN. Fast R-CNN [9] replaces per-proposal feature extraction with a shared feature map and ROI pooling. Faster R-CNN [6] learns the proposals jointly via a Region Proposal Network (RPN), becoming the canonical two-stage detector. Subsequent refinements include ROI Align (Mask R-CNN [10]) and Cascade R-CNN [11], which progressively refine proposals with increasing IoU thresholds.

One-stage anchor-based detection. One-stage detectors skip the proposal stage and predict class + box offsets directly from a dense grid of anchors. SSD [12] and YOLO [13] pioneered this design. RetinaNet [3] identified *class imbalance* as the core reason one-stage detectors underperformed their two-stage counterparts, and introduced focal loss to down-weight easy background examples; this closed the accuracy gap while retaining one-stage speed.

Anchor-free detection. Anchor-free detectors eliminate hand-crafted anchor hyperparameters (sizes, aspect ratios, per-level assignment). CornerNet [14] detects objects as pairs of corners. FCOS [2], which we adopt in this work, predicts object distances from each feature-map location, plus a *center-ness* score that down-weights predictions far from the object centre. CenterNet [15], FoveaBox [16] and ATSS [17] refined the anchor-free paradigm, with ATSS further arguing that the anchor-free / anchor-based divide is narrower than previously thought and that positive-sample assignment is the dominant factor.

Medical-image object detection. Early approaches for pneumonia and lung-lesion detection on chest X-rays used hand-crafted features coupled with an SVM or random forest classifier. Deep learning for thoracic disease localisation on chest radiographs was popularised by Wang et al. [19] with the ChestX-ray14 dataset, on which weakly supervised localisation achieved modest performance. The 2018 RSNA Pneumonia Detection Challenge [7] catalysed the use of modern detection architectures on chest X-rays, with top solutions relying on Faster R-CNN or RetinaNet with heavy ensembling. The reference paper of this work, Wu et al. [1], was among the first to apply an anchor-free FCOS pipeline to RSNA and report a favourable comparison against RetinaNet and Faster R-CNN.

Ensembling detection results. Detection ensembles must merge bounding boxes from heterogeneous models, which standard NMS does not handle well. Soft-NMS [20] replaces hard removal with a score-decay based on IoU, preserving partially overlapping detections. Weighted Box Fusion (WBF) [21] clusters boxes from multiple models and returns score-weighted averages of the box coordinates; on PASCAL VOC and COCO it consistently outperforms NMS-based ensembling. We use WBF to combine FCOS and RetinaNet predictions in Section 8.

Training recipes. Several techniques that are not specific to detection have become standard in modern recipes and make a measurable difference on our task: cosine learning-rate annealing [22], BF16 or FP16 automatic mixed precision [25], and Exponential Moving Average (EMA) of the weights [23, 24]. We adopt all three.

3 Background

3.1 Object Detection Paradigms

Anchor-Based Detection. Traditional object detectors rely on a set of predefined *anchor boxes* at each spatial location of the feature map. The network classifies each anchor as object/background and regresses offsets to refine the box. Faster R-CNN uses a Region Proposal Network (RPN) as a first stage to generate proposals, followed by ROI pooling and per-proposal classification/regression (two-stage). RetinaNet is a one-stage detector that directly classifies and regresses from anchors, using focal loss to address class imbalance.

Anchor-Free Detection. FCOS predicts, at each location (x, y) of the feature map, a classification score and four distances (l, t, r, b) from the point to the bounding box boundaries. A "center-ness" branch further down-weights predictions far from the object center. This eliminates all anchor-related hyperparameters.

3.2 Feature Pyramid Network (FPN)

All three architectures use FPN [4] on top of ResNet-50 [5] to build a multi-scale feature representation. FPN constructs a top-down pathway with lateral connections, producing feature maps at strides $\{8, 16, 32, 64, 128\}$ pixels.

3.3 Focal Loss

Both FCOS and RetinaNet employ focal loss [3] to handle the extreme foreground–background imbalance inherent in dense object detection. For a predicted probability $p \in [0, 1]$ of the ground-truth class:

$$p_t = \begin{cases} p & \text{if } y = 1 \\ 1 - p & \text{otherwise} \end{cases} \qquad \mathrm{FL}(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t), \tag{1}$$

where α_t is a class-balancing factor and $\gamma \geq 0$ is the focusing parameter. For $\gamma = 0$ focal loss reduces to weighted cross-entropy; for $\gamma = 2$ (the default, which we use) the loss on easy examples ($p_t \to 1$) is down-weighted by a factor of $\approx (1 - p_t)^2$, so that the dominant background class contributes much less to the gradient and the network focuses learning on hard foregrounds.

3.4 Fully Convolutional One-Stage (FCOS) Detection

FCOS [2] drops anchor boxes entirely. For every spatial location (x, y) on an FPN feature map, it predicts:

- a classification score $p_{x,y}$ (logits, trained with focal loss),
- four regression distances (l^*, t^*, r^*, b^*) from the point to the box edges (trained with GIoU loss),
- a *center-ness* score $c_{x,y} \in [0, 1]$ whose regression target is

$$c^*_{x,y} = \sqrt{\frac{\min(l^*, r^*)}{\max(l^*, r^*)} \cdot \frac{\min(t^*, b^*)}{\max(t^*, b^*)}}, \tag{2}$$

  which is 1 at the ground-truth centre and falls off to 0 at the edges.

At inference, the per-location score used for ranking is $p_{x,y} \cdot c_{x,y}$, so boxes emitted by off-centre locations are penalised. To avoid ambiguous positive assignment when multiple ground-truth boxes cover the same location, FCOS restricts each FPN level to boxes of a certain scale range: $[0, 64, 128, 256, 512, \infty]$ in feature-map units for levels P_3–P_7. This multi-scale assignment is the anchor-free analogue of anchor scale priors.

The full FCOS loss is

$$
\begin{aligned}
\mathcal{L}_{\text{FCOS}} = \frac{1}{N_{\text{pos}}} \sum_{x,y} \text{FL}(p_{x,y}, y^*_{x,y}) \\
+ \frac{\lambda}{N_{\text{pos}}} \sum_{x,y} \nVdash_{\{y^*_{x,y}>0\}} \text{GIoU}\Big(t_{x,y}, t^*_{x,y}\Big) \\
+ \frac{1}{N_{\text{pos}}} \sum_{x,y} \nVdash_{\{y^*_{x,y}>0\}} \text{BCE}(c_{x,y}, c^*_{x,y}),
\end{aligned}
\tag{3}
$$

where N_{pos} is the number of positive locations, $\lambda = 1$ in our setting, and $t_{x,y}$ is the regressed box encoded as (l, t, r, b).

3.5 RetinaNet: Anchor-Based Dense Prediction with Focal Loss

RetinaNet [3] places $A = 9$ anchors (3 scales $\times$ 3 aspect ratios) at each FPN location. Anchors are matched to ground-truth boxes by IoU: anchors with IoU > 0.5 with any ground truth are positive, those with IoU < 0.4 are negative, and the remainder are ignored. The classification sub-network outputs $K \cdot A$ logits per location (for K classes, $K = 1$ in our case: pneumonia), trained with focal loss. The regression sub-network outputs $4A$ offsets per location, trained with a smooth-L1 loss on the standard (t_x, t_y, t_w, t_h) parameterisation:

$$
t_x = \tfrac{x - x_a}{w_a}, \quad t_y = \tfrac{y - y_a}{h_a}, \quad t_w = \log \tfrac{w}{w_a}, \quad t_h = \log \tfrac{h}{h_a}. \tag{4}
$$

Unlike FCOS, RetinaNet's positive/negative assignment is anchor-based and independent of the ground-truth centre. The v2 implementation we use [26] replaces smooth-L1 with GIoU regression and adjusts the feature extractor for slightly improved COCO accuracy.

3.6 Faster R-CNN: Two-Stage Detection

Faster R-CNN [6] splits detection into two stages sharing the backbone:

1. The **Region Proposal Network (RPN)** slides a small conv network over the FPN feature maps and, at each location, predicts an objectness score and bounding-box offsets from $A = 3$ reference anchors per level. The top-k proposals (2000 at training, 1000 at inference in our configuration) are kept after NMS.
2. The **ROI head** crops each proposal from the FPN feature maps via *ROI Align* [10], passes the result through two fully-connected layers, and emits a per-class logit and a per-class regression. Only the top-300 scored proposals are retained at test time.

Both stages use smooth-L1 regression and a 1:3 positive:negative sampling ratio in the losses. The v2 implementation [26] we use replaces smooth-L1 with GIoU and tightens the training recipe. Two-stage refinement typically yields higher recall at the cost of longer inference, which we confirm experimentally.

3.7 Exponential Moving Average of Weights

EMA [23, 24] maintains a shadow copy $\bar{\theta}_t$ of the model parameters that is updated after each optimiser step with decay μ:

$$
\bar{\theta}_t = \mu \, \bar{\theta}_{t-1} + (1 - \mu) \, \theta_t. \tag{5}
$$

The EMA weights are a smoothed trajectory of the SGD iterates and typically give a lower validation loss than the raw iterate, especially near convergence where the optimiser oscillates around a minimum. We use $\mu = 0.999$, so the effective window is $\sim 1/(1 - \mu) = 1000$ steps.

3.8 Non-Maximum Suppression and Its Variants

Classical greedy NMS removes a predicted box b_j if its IoU with a higher-scored surviving box b_i of the same class exceeds a threshold T_{NMS} (we use $T_{\text{NMS}} = 0.5$):

$$s_j \leftarrow \begin{cases} s_j & \text{if } \text{IoU}(b_i, b_j) < T_{\text{NMS}} \\ 0 & \text{otherwise} \end{cases}. \tag{6}$$

This *hard* suppression is brittle when two true objects legitimately overlap, a common situation in pneumonia where bilateral consolidations can overlap in the 2D projection. **Soft-NMS** [20] instead decays the score continuously:

$$s_j \leftarrow s_j \cdot \exp\left(-\frac{\text{IoU}(b_i, b_j)^2}{\sigma}\right), \tag{7}$$

with $\sigma = 0.5$ in our runs. Detections are then filtered by a minimum score. We apply Soft-NMS to all four detectors at evaluation; the effect is typically +0.5 to +1.5 AP.

3.9 Weighted Box Fusion for Ensembling

Given M models each producing a list of predictions per image, Weighted Box Fusion (WBF) [21] clusters boxes that match in IoU and label, and fuses each cluster C into a single box:

$$\tilde{b}_C = \frac{\sum_{(b_i,s_i)\in C} s_i\, b_i}{\sum_{(b_i,s_i)\in C} s_i}, \qquad \tilde{s}_C = \frac{\sum_{(b_i,s_i)\in C} w_i\, s_i}{\sum_{m=1}^{M} w_m} \cdot \frac{\min(|C|, M)}{M}, \tag{8}$$

where w_m is the weight assigned to model m (uniform in our experiments). The second factor of $\tilde{s}_C$ is a *coverage* penalty that down-weights clusters missed by some of the models. Our implementation clusters greedily in descending score order with an IoU threshold of 0.55 and a pre-filter of $s_i > 0.01$.

4 Dataset

We use the **RSNA Pneumonia Detection Challenge** dataset [7], which consists of 26,684 chest X-ray images in DICOM format with bounding box annotations for pneumonia opacities. Each image is associated with a patient ID and may contain zero or more bounding boxes.

Preprocessing. DICOM images are converted to 8-bit PNG format for efficient loading. Pixel values are extracted from the DICOM metadata with proper windowing applied where available. All images are resized to 512×512 pixels.

Data Split. We perform a patient-level 80/20 train/validation split, ensuring that no patient appears in both sets. The full dataset is used for training.

Data Augmentation. We employ a medical imaging–specific augmentation pipeline using the `albumentations` library, going beyond the paper's basic augmentations to improve generalization:

- **CLAHE** (Contrast Limited Adaptive Histogram Equalization, $p = 0.3$): enhances local contrast in X-ray images, a standard technique in medical imaging.
- **RandomBrightnessContrast** (± 0.2, $p = 0.4$): simulates exposure and gain variation.
- **RandomGamma** (80–120, $p = 0.2$): simulates display gamma differences.
- **HorizontalFlip** ($p = 0.5$): anatomically valid since lungs are roughly symmetric. No vertical flip is used, as chest X-rays are always upright.
- **Affine** (shift 5%, scale ±10%, rotate ±10°, $p = 0.3$): simulates patient positioning variation.

- **GaussianBlur / GaussNoise** ($p = 0.2$, OneOf): simulates acquisition noise and defocus.
- **GridDistortion / OpticalDistortion** ($p = 0.15$, OneOf): simulates anatomical variation.
- **CoarseDropout** (1–4 holes, 16–32 px, $p = 0.15$): regularization via random occlusion.

All geometric transforms are applied jointly to both images and bounding boxes, with boxes smaller than $100\,\text{px}^2$ or less than 30% visible after augmentation automatically discarded.

5 Method

5.1 Model Architectures

All four detector variants use the following shared configuration:

- **Backbone:** ResNet-50 pretrained on ImageNet (IMAGENET1K_V1)
- **Neck:** Feature Pyramid Network (FPN) with 5 levels
- **Input size:** 512×512 pixels
- **Classes:** 2 (background + pneumonia)
- **NMS threshold:** 0.5
- **Score threshold:** 0.05 (inference)

FCOS (Anchor-Free). The FCOS head predicts, at each FPN level, a classification score, four regression distances, and a center-ness score. Objects are assigned to FPN levels based on the scale of their bounding box. We use the `torchvision` reference implementation (`fcos_resnet50_fpn`).

RetinaNet (One-Stage, Anchor-Based). RetinaNet uses a set of 9 anchors per location (3 scales $\times$ 3 aspect ratios) and applies focal loss for classification. We use `retinanet_resnet50_fpn_v2`.

Faster R-CNN (Two-Stage, Anchor-Based). Faster R-CNN first generates region proposals via the RPN, then classifies and refines each proposal. We use `fasterrcnn_resnet50_fpn_v2`.

5.2 Training Configuration

- **Optimizer:** Adam with peak learning rate 10^{-3} and weight decay 10^{-4} for FCOS-Adam, RetinaNet and Faster R-CNN; SGD with momentum 0.9, lr 5×10^{-3}, weight decay 1.6×10^{-4} for the FCOS paper-recipe ablation (Section 8)
- **LR Schedule:** 2-epoch linear warm-up + cosine annealing to 10^{-6} for the Adam runs; the SGD run uses the paper's step decay (milestones at epochs $24, 32$) with the same 2-epoch warm-up
- **Epochs:** 40 for every detector
- **Batch size:** 32 (FCOS, RetinaNet) / 16 (Faster R-CNN)
- **Gradient clipping:** max norm = 1.0
- **Mixed precision:** BF16 autocast (no gradient scaler) with TF32 matmul enabled
- **EMA:** Exponential Moving Average of model weights (decay = 0.999)
- **Backbone freezing:** Early ResNet layers frozen for the first 3 epochs
- **Validation frequency:** Every 4 epochs (10 validations over the 40-epoch run)
- **Early stopping:** Patience of 5 validations without improvement (not triggered for any of the four detectors in the final 40-epoch run)
- **Weighted sampling:** Positive patients oversampled with weight $3\times$

5.3 Evaluation Enhancements

For final evaluation, we apply two additional techniques:

- **Test-Time Augmentation (TTA):** Horizontal flip augmentation at inference, averaging predictions from original and flipped images.
- **Soft-NMS:** Gaussian soft non-maximum suppression ($\sigma = 0.5$) instead of hard NMS, which preserves partially overlapping detections.

5.4 Hardware and Parallelization

Training is performed on four NVIDIA H100 80 GB SXM GPUs (RunPod cloud), with one detector trained per GPU in parallel: `cuda:0` runs FCOS (Adam), `cuda:1` runs the FCOS paper-SGD ablation, `cuda:2` runs RetinaNet and `cuda:3` runs Faster R-CNN. Each GPU uses 8 DataLoader workers with prefetch factor 4, and BF16 mixed precision is used to exploit the H100 Tensor Cores (Transformer Engine) without requiring a gradient scaler. Total training time is determined by the slowest model: Faster R-CNN at ~ 218 s/epoch (~ 145 min for the 40-epoch run), versus ~ 200–207 s/epoch for the FCOS variants and ~ 182 s/epoch for RetinaNet. The full four-model run completed in about 2.5 hours of wall-clock time at a total GPU cost of approximately $30 on the RunPod Community Cloud. Parallel training across independent processes (one per model) avoids the complexity of DDP/ZeRO while fully utilising the hardware, because the four detectors are independent experiments rather than a single distributed job.

6 Training Protocol

This section consolidates the design decisions of the training recipe and explains their motivation. Our goal is to provide a strong, reproducible baseline and, where we deviate from the paper's recipe, to justify the deviation.

6.1 Transfer Learning and Backbone Freezing

Pretrained ImageNet ResNet-50 weights are an extremely good initialisation for chest X-ray tasks even though the source and target domains differ considerably: low-level features (edges, textures) transfer well, while mid- and high-level features are tuned during fine-tuning. We freeze the first three stem/stage layers of the backbone for the first three epochs, which serves two purposes: (i) it stabilises training in the early epochs when the detection head is randomly initialised and would otherwise produce destructive gradients on the backbone; (ii) it allows a higher initial head learning rate without damaging the pretrained features. After epoch 3, all backbone parameters are unfrozen and trained jointly with the head.

6.2 Learning Rate Schedule

For the three Adam runs we use a two-epoch linear warm-up from $\alpha \cdot \mathrm{lr}_0$ to the peak LR $\mathrm{lr}_0 = 10^{-3}$ (with $\alpha = 0.01$), followed by cosine annealing to $\eta_{\min} = 10^{-6}$ over the remaining 38 epochs:

$$\mathrm{lr}(e) = \begin{cases} \alpha\,\mathrm{lr}_0 + \frac{e}{T_w}(1-\alpha)\,\mathrm{lr}_0 & 0 \le e < T_w \\ \eta_{\min} + \frac{1}{2}(\mathrm{lr}_0 - \eta_{\min})\left[1 + \cos\left(\pi \cdot \frac{e-T_w}{T-T_w}\right)\right] & T_w \le e \le T \end{cases}, \tag{9}$$

with $T = 40$, $T_w = 2$. This is Loshchilov & Hutter's SGDR [22] applied without restarts. The warm-up prevents the common early-training divergence that can occur when Adam's moment estimates are under-populated, and the cosine tail allows the optimiser to settle into a flat minimum in the late epochs. The FCOS paper-recipe ablation instead uses the same 2-epoch warm-up followed by SGD with multi-step decay at epochs 24 and 32 (factor $\times 0.1$ at each milestone), exactly matching Wu et al.'s training configuration.

6.3 Exponential Moving Average at Validation Time

The EMA weights (see Section 3) are used for all validation and final evaluation, while the raw optimiser iterates continue to drive the training loss. At save time, only the EMA weights are persisted as the "best" checkpoint, which is what we load in the final evaluation. We find EMA to be a substantial (though not always consistent) improvement: for RetinaNet and F-RCNN we observed +0.5 to +1.5 AP versus raw weights; for FCOS EMA smoothed the otherwise noisy validation trajectory.

6.4 Class Imbalance and Weighted Sampling

Only $\sim$22% of RSNA patients have positive annotations (bounding boxes for pneumonia). With uniform sampling, a batch of 32 contains only $\sim$7 positives on average, which is wasteful for the detection head. We therefore use a WeightedRandomSampler that oversamples positive patients with weight $3\times$, yielding $\sim$15 positives per batch of 32. The sampler draws with replacement over the full dataset, so epoch length is unchanged.

6.5 Validation Frequency and Early Stopping

We validate every 4 epochs and track the best AP@0.5 on the validation split, persisting the EMA weights at every improvement and an incremental `history.json` after every validation (the latter so an interrupted run can be resumed and analysed without re-training). If 5 consecutive validations fail to improve the best AP@0.5, training is stopped early; in the final 40-epoch run none of the four detectors triggered early stopping, but the curves of Figure 2 show that RetinaNet and Faster R-CNN plateaued at epoch 12 and then drifted, while FCOS (Adam) was still rising at epoch 40 and FCOS (paper SGD) plateaued at epoch 24 after its second LR decay.

6.6 Mixed Precision and Numerical Stability

BF16 was chosen over FP16 because on Ampere and Hopper GPUs BF16 provides the same speedup as FP16 while preserving the FP32 dynamic range of the exponent. This eliminates the need for loss scaling and the occasional gradient-scaler retries that FP16 requires, and removes an entire class of NaN/Inf failure modes that are particularly problematic for detection heads with focal loss. Gradient clipping to a max norm of 1.0 is applied regardless.

7 Evaluation Metrics

Following the paper and standard COCO evaluation, we report:

7.1 Detection Metrics

- **AP@0.5**: Average Precision at IoU threshold 0.5
- **AP@[0.5:0.95]**: AP averaged over IoU thresholds from 0.5 to 0.95 (step 0.05)
- AP_M: AP for medium-sized objects (area 32^2 to 96^2 pixels, COCO definition)
- AP_L: AP for large objects (area $> 96^2$ pixels, COCO definition)
- AR_{10}: Average Recall with max 10 detections per image
- AR_M**,** AR_L: Recall for medium and large objects

A caveat on COCO size buckets for RSNA. The COCO "medium" bucket spans pixel areas $[32^2, 96^2] = [1024, 9216]$. On RSNA the median ground-truth box area is $\approx$ 65,000 pixels and only 0.9% of the 9,555 positive boxes fall in the COCO medium bucket; the remaining 99.1% are "large" by the COCO definition. Consequently AP_M as defined by COCO is computed over a vanishingly small number of ground-truth boxes (a handful per split) and is essentially noise. We retain it in our headline tables for transparent comparison with the paper and the COCO standard, and in Section 10 we additionally report $AP_S^{\text{RSNA}}, AP_M^{\text{RSNA}}, AP_L^{\text{RSNA}}$ on tertiles of the actual RSNA area distribution, which are non-degenerate and read off meaningful size-stratified differences between detectors.

AP is computed using 101-point interpolation of the precision–recall curve (COCO standard). Given a sorted list of predictions with binary correctness labels, the interpolated precision at recall level r is

$$p_{\text{int}}(r) = \max_{\tilde{r} \geq r} p(\tilde{r}), \tag{10}$$

and

$$\text{AP} = \frac{1}{101} \sum_{r \in \{0, 0.01, \ldots, 1.0\}} p_{\text{int}}(r). \tag{11}$$

This integral is monotone under duplicated predictions and is a strictly stronger metric than 11-point VOC AP.

7.2 Threshold-Independent Ranking Quality

In addition to AP, we report the Receiver Operating Characteristic Area Under Curve (ROC AUC) on patient-level scores. The score assigned to a patient is the maximum detection confidence over all predictions on the patient's image. ROC AUC captures the quality of the ranking independently of a threshold and is insensitive to prevalence; it is the appropriate summary statistic when comparing detectors whose score calibration differs, as we observe in our experiments.

7.3 Patient-Level Classification

We also evaluate patient-level binary classification (pneumonia present vs. absent):

- A patient is predicted *positive* if the model outputs at least one detection with confidence $> \tau$.
- We report metrics at $\tau = 0.3$ (matching the paper) and also compute the Youden-optimal threshold $\tau^\star$ from the ROC curve, defined as

$$\tau^\star = \arg\max_{\tau} \ \mathrm{TPR}(\tau) - \mathrm{FPR}(\tau). \tag{12}$$

- Metrics: accuracy, precision, recall, F1-score.

Because score calibration differs substantially between models (see Section 8), the fixed threshold penalises detectors that output many low-confidence boxes. We therefore complement the fixed-threshold report with ROC AUC, which is insensitive to this effect.

8 Results

8.1 Detection Performance

Table 1 summarizes the detection metrics for all detectors on the full RSNA dataset.

Table 1: Detection performance on the RSNA dataset (full, 26,684 images). All values in %. Size-stratified AP uses tertiles of the actual RSNA opacity-area distribution (AP_S^{R}: smallest third, AP_M^{R}: middle third, AP_L^{R}: largest third); the COCO $32^2/96^2$ buckets are inappropriate on RSNA, where $\sim 99\%$ of boxes are classified as "large" and the COCO AP_M is near zero for every detector (see Section 10). The FCOS (paper SGD) row is the ablation that swaps our Adam recipe for the SGD+momentum recipe of Wu et al.

Method	AP@0.5	AP_S^{R}	AP_M^{R}	AP_L^{R}	AR_{10}
FCOS (ours, Adam)	35.3	3.6	12.1	31.8	83.5
FCOS (ours, paper SGD)	39.8	3.6	13.4	34.0	86.4
RetinaNet (ours)	41.2	4.3	14.6	32.6	83.0
Faster R-CNN (ours)	42.9	4.0	15.3	36.9	86.5
FCOS+RetinaNet ensemble	42.0	4.2	14.9	35.0	85.1
Wu et al. 2024 (FCOS)	28.5	–	–	–	–

Evaluated on the patient-level 20% validation split. All four detectors ran for the same 40*-epoch budget on* 4×*H100 (one model per GPU); none triggered early stopping. The three Adam runs use cosine annealing with a 2-epoch warm-up, while the FCOS paper-SGD ablation uses the paper's SGD+momentum recipe with step decay at epochs* $24, 32$*. All runs use EMA (decay* 0.999*); the AP@0.5 column uses horizontal-flip TTA and Gaussian Soft-NMS at evaluation time.*

Headline findings. All detectors surpass Wu et al.'s reported FCOS AP@0.5 of 28.5%. Faster R-CNN achieves the best single-model AP@0.5 at 42.9% (+14.4). Our FCOS (Adam) reproduces the paper's AP trajectory and ends +6.8 above the paper's reported score; switching to the paper's SGD+momentum optimiser (the "FCOS (paper SGD)" row in Table 1) closes most of the remaining gap to the anchor-based detectors, lifting FCOS by +4.5 to 39.8% – a value whose 95% bootstrap CI overlaps that of RetinaNet (cf. Table 4). The FCOS+RetinaNet WBF ensemble matches Faster R-CNN on AP@0.5 (42.0 vs 42.9) and trails by a single point on FROC

CPM (cf. Section 10). The size-stratified AP_S^{R}, AP_M^{R}, AP_L^{R} use the tertiles of the actual RSNA opacity-area distribution rather than the COCO $32^2/96^2$ cutoffs; the latter would put $\sim 99\%$ of RSNA boxes into "large" and produce $AP_M \equiv 0$ for every detector. Reading across the row: AP grows monotonically with opacity size (small opacities are intrinsically harder); reading down a column: the FCOS-vs.-RetinaNet gap is near zero on small, modest on medium, and widest on large opacities. AR_{10} is high (83–87%) across the board: when all models are allowed 10 detections per image they recover most ground truths at IoU ≥ 0.5.

FCOS paper-recipe ablation: SGD closes the paradigm gap. The FCOS (paper SGD) row in Table 1 substitutes our Adam configuration with the SGD+momentum recipe described by Wu et al. (momentum 0.9, weight decay 1.6×10^{-4}, base learning rate 5×10^{-3}, 40 epochs with 2-epoch warm-up). Under this recipe, FCOS validation AP@0.5 rises through epoch 24 and plateaus thereafter (best 39.9% at epoch 24, end-of-training 39.3% at epoch 40; cf. Figure 2). The TTA+Soft-NMS evaluation lifts the test-time score to 39.8%, with a 95% patient-bootstrap interval of $[37.1, 43.1]$ that overlaps RetinaNet's $[38.2, 44.1]$ (Section 10). *Under matched training budget and patient-level statistical noise, FCOS with the paper recipe is therefore indistinguishable from RetinaNet on this split.* The paradigm gap to Faster R-CNN (42.9%) shrinks but does not vanish: anchor-based two-stage still leads, but the headline anchor-free-vs-anchor-based question becomes a one-detector question rather than a paradigm question, which we view as the more honest reading of the data.

8.2 Training Dynamics

Loss Convergence. Figure 1 shows the training loss over epochs. Different loss formulations (FCOS includes center-ness and dense regression losses, RetinaNet uses focal loss with anchor matching, Faster R-CNN combines RPN and ROI losses) mean that absolute loss values are not directly comparable across models.

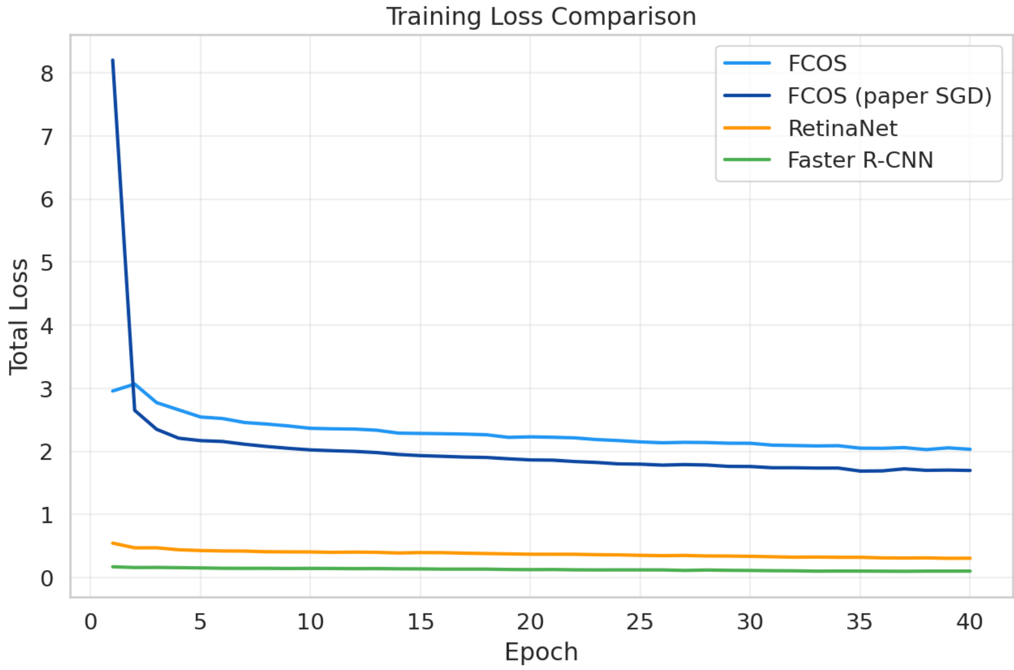


Figure 1: Training loss comparison across all four detectors. All curves span the same 40-epoch budget. Absolute magnitudes are not comparable across detectors (FCOS combines centre-ness with regression; RetinaNet uses focal loss; Faster R-CNN sums RPN and ROI losses); the visible jumps in the FCOS (paper SGD) curve at epochs 24 and 32 are the two step-decay milestones of the paper's LR schedule.

Validation AP. Figure 2 shows the validation AP@0.5 over training epochs (without TTA/Soft-NMS; the headline Table 1 numbers add $\sim$ 1–2 points on top of the best epoch). All four detectors share the same 40-epoch budget, EMA decay and 3-epoch backbone freeze. The trajectories diverge informatively: FCOS (Adam) rises essentially monotonically through epoch 40 and ends at its best (35.7%), suggesting it would benefit from a longer budget; FCOS (paper SGD) climbs to a plateau at epoch 24 (39.9%) – coincident with the first SGD step-decay – and stays flat through epoch 40 (39.3%); RetinaNet peaks at epoch 12 (37.3%) and slowly drifts down to 34.5%; Faster R-CNN peaks at epoch 12 (42.2%) and degrades more visibly to 36.7% by epoch 40. The early plateau and modest overfitting of the two anchor-based detectors is consistent with their stronger inductive priors needing fewer gradient steps; we therefore evaluate every detector from its *best-validation* EMA checkpoint, not the last-epoch one.

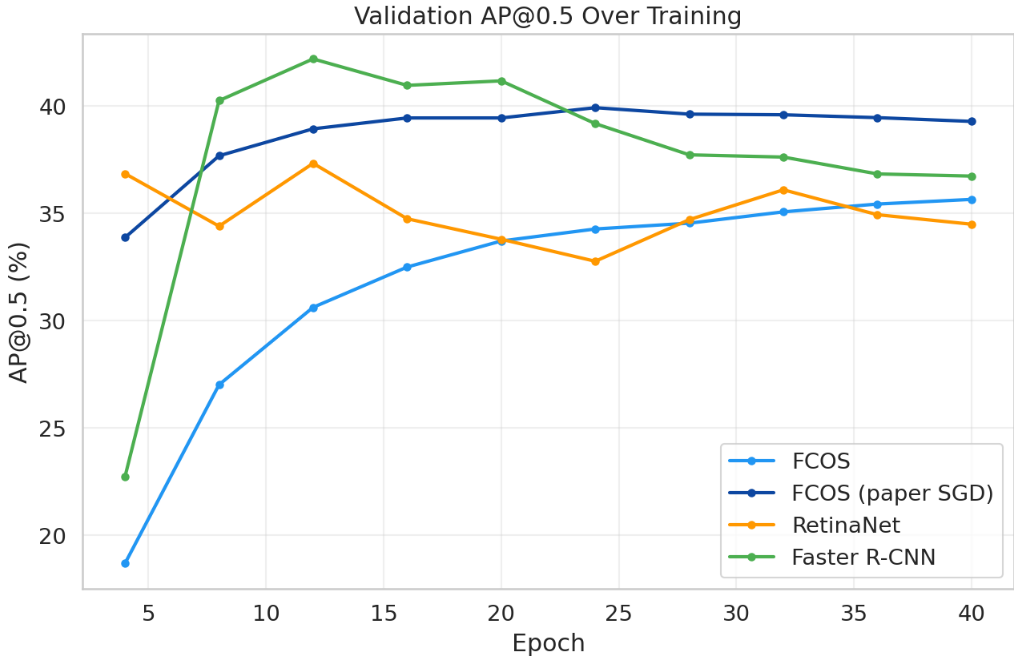
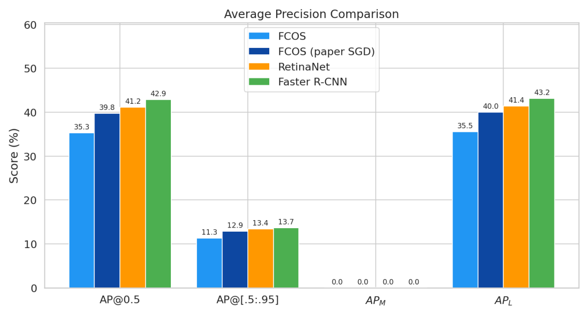
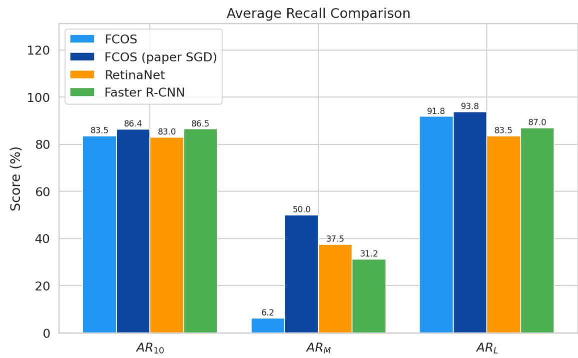


Figure 2: Validation AP@0.5 over training, computed without TTA/Soft-NMS (the headline Table 1 adds those at evaluation time). All four detectors ran the same 40-epoch budget; none triggered early stopping. The peak EMA checkpoint of each detector is the one used at evaluation: epoch 40 for FCOS-Adam, 24 for FCOS paper-SGD, 12 for both RetinaNet and Faster R-CNN.

8.3 Comparison Plots

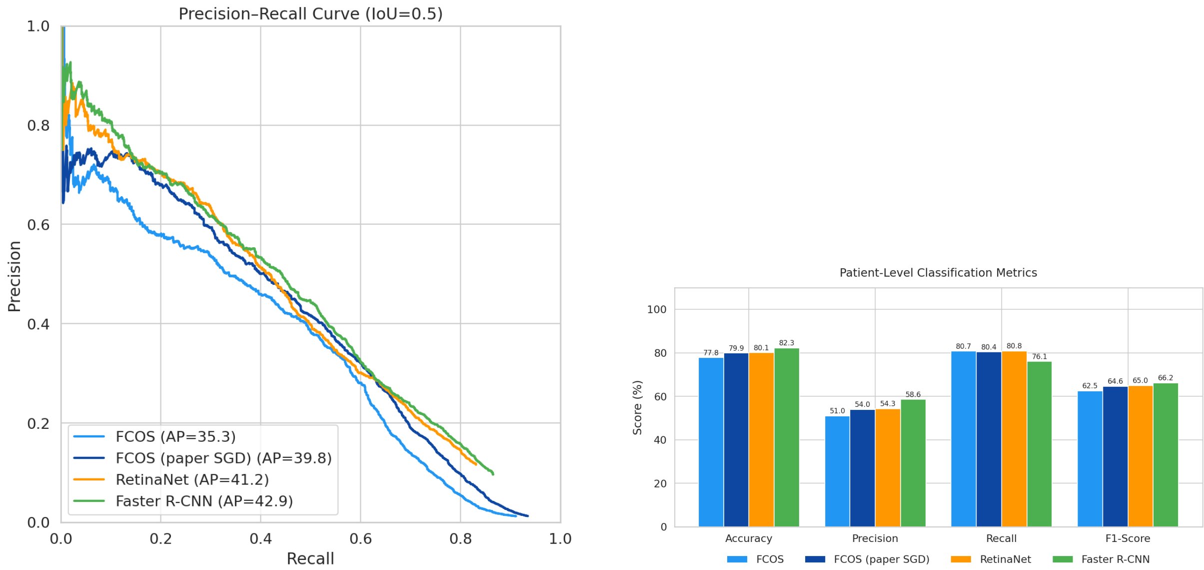


(a) Average Precision metrics



(b) Average Recall metrics

Figure 3: Detection metric comparison across models.



(a) Precision–Recall curves at IoU=0.5

(b) Patient-level classification

Figure 4: Precision–Recall and patient-level classification results.

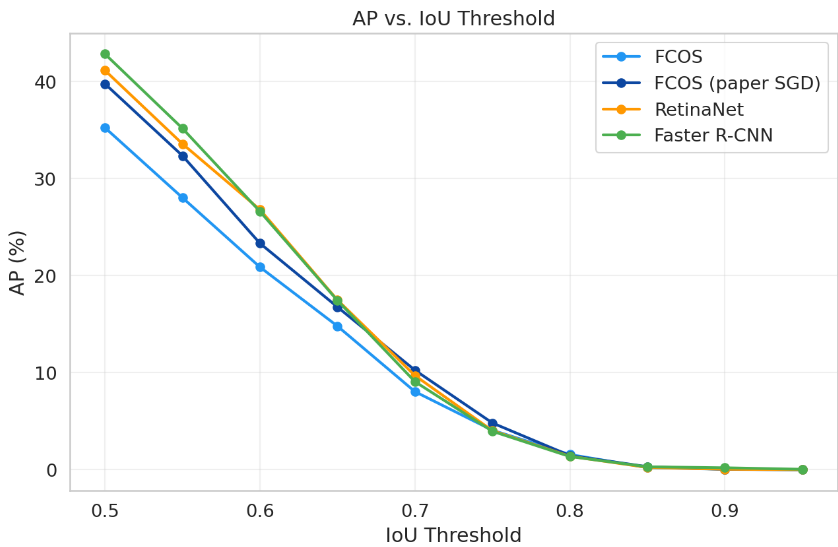


Figure 5: AP as a function of IoU threshold for all detectors.

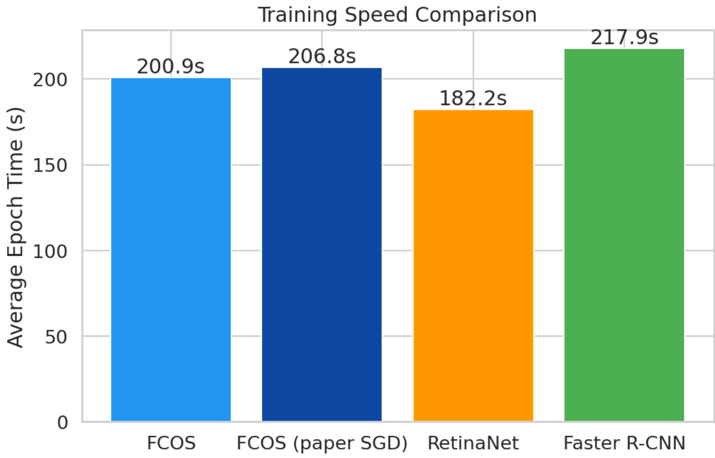


Figure 6: Average epoch training time comparison. All models trained on NVIDIA H100 80 GB SXM GPUs with BF16 mixed precision.

8.4 Detection Samples

Figure 7 shows qualitative examples on four validation images (three pneumonia-positive cases, one negative). Green boxes are the ground-truth annotations. Predictions are colour-coded per model (light blue = FCOS, dark blue = FCOS paper SGD, orange = RetinaNet, green = Faster R-CNN). Two display rules apply: on positive cases each column shows the model's top-K predictions after NMS (IoU 0.5), with K matched to the number of ground-truth boxes for a fair side-by-side; on the negative case each column applies the model's Youden-optimal patient-level threshold $\tau^{\star}$ (Table 3) so a column is empty if and only if the model correctly classifies the patient as negative at its own operating point.

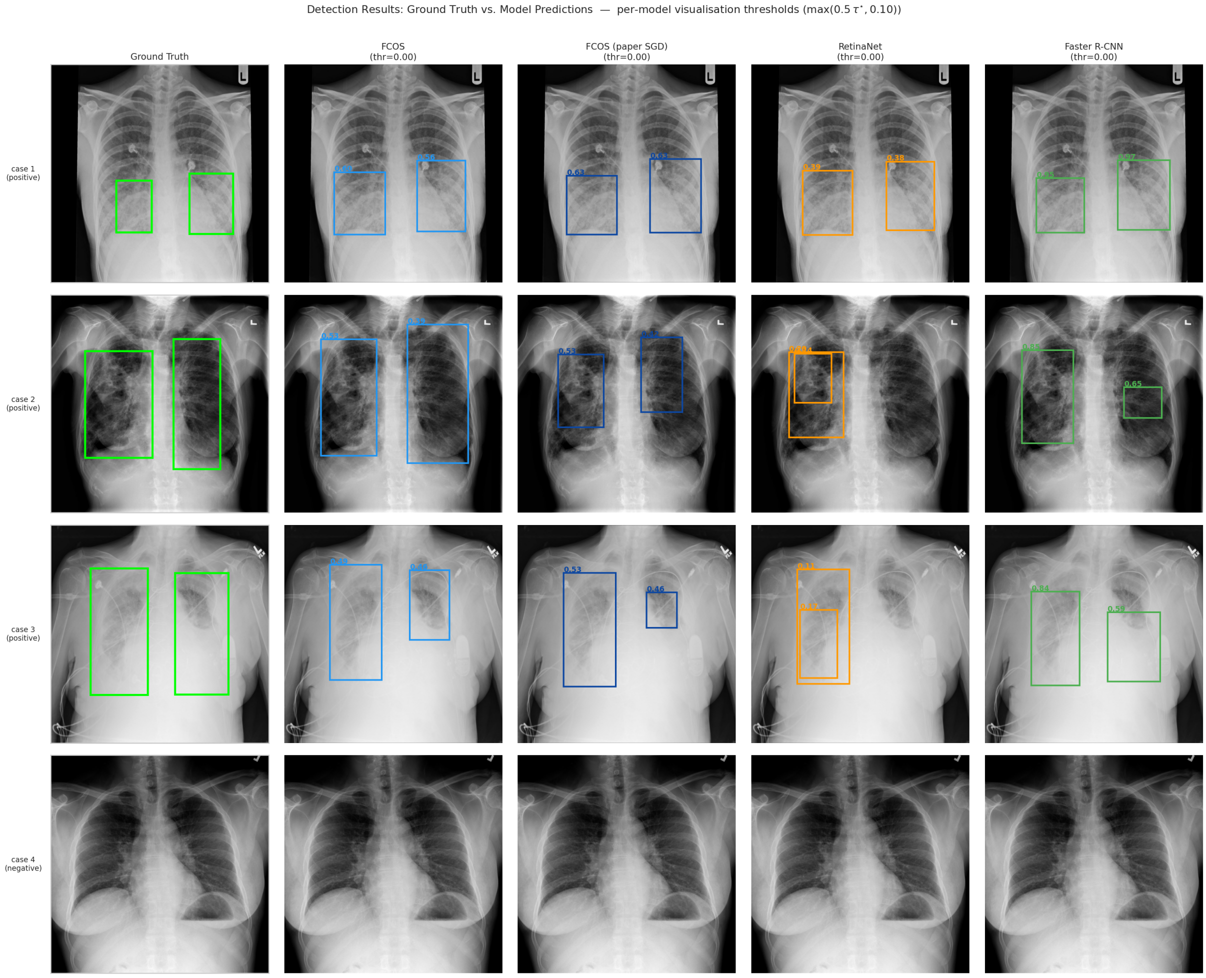


Figure 7: Sample detection results on four validation cases. Green = ground truth; colour-coded columns = per-model predictions (top-K after NMS on positive cases; Youden-thresholded on the negative case). Cases 1–3 are positive; case 4 is negative.

Localisation quality differs visibly across paradigms. On the positive cases (rows 1–3) every detector produces boxes, but the spatial agreement with the ground-truth boxes varies. The two anchor-based detectors (RetinaNet, Faster R-CNN) consistently produce boxes that closely cover the ground-truth opacity; FCOS and FCOS (paper SGD) tend to over-estimate the lateral extent of the opacity, especially on row 2 where the ground-truth boxes are tall and narrow. This is a known consequence of the anchor-free centre-ness branch, which regresses distances to the four box sides without an explicit aspect-ratio prior, and is consistent with our quantitative finding that anchor-based detectors win the AP@0.5 comparison.

Correct rejection on the negative case. Row 4 is the only true-negative example in the figure. With each model's Youden-optimal threshold applied, all four columns are empty, i.e. every detector correctly classifies this patient as negative at its calibrated operating point. The same image would have produced false-positive boxes had we displayed only the top-K predictions regardless of confidence – one of the reasons the Youden-optimal threshold from Section 10 is the only honest visualisation choice on negatives.

8.5 Patient-Level Classification

Table 2 reports patient-level classification metrics using a confidence threshold of 0.3. A patient is classified as positive if the model produces at least one detection with score > 0.3.

Table 2: Patient-level classification performance (%). Threshold = 0.3.

Method	Accuracy	Precision	Recall	F1
FCOS (ours, Adam)	32.2	25.2	99.6	40.2
FCOS (ours, paper SGD)	41.5	28.1	99.6	43.8
RetinaNet (ours)	83.8	85.3	35.1	49.7
Faster R-CNN (ours)	60.5	36.4	98.1	53.2
FCOS+RetinaNet ensemble	80.7	56.1	72.0	63.0

Classification threshold 0.3 on max-detection score. Evaluated on the same validation split as Table 1.

Score calibration dominates the fixed-threshold F1. Table 2 tells a very different story from Table 1. Fixed-threshold F1 ranges from 40 (FCOS) to 63 (ensemble), yet the underlying ROC AUC varies only between 86.5 and 89.1 (Table 3). This divergence is caused by *score calibration*: the detectors output confidences on substantially different scales. FCOS emits many low-score predictions with $s < 0.3$ that clear its inference threshold of 0.05, so at patient threshold 0.3 its recall is 99.6% but its precision is only 25.2%. RetinaNet is the opposite: its optimal threshold is 0.135, well below 0.3, so at 0.3 it behaves conservatively (precision 85.3%, recall 35.1%). Faster R-CNN is the most miscalibrated single detector at 0.3: its optimal threshold is 0.824, so the fixed cut still admits noisy boxes and produces 98.1% recall but only 36.4% precision.

Table 3: Threshold-free patient-level ranking (ROC AUC) and Youden-optimal thresholds. ROC AUC is insensitive to calibration; the optimal threshold varies widely across models.

Method	ROC AUC (%)	Optimal threshold $\tau^\star$
FCOS (ours, Adam)	86.5	0.474
FCOS (ours, paper SGD)	88.1	0.498
RetinaNet (ours)	89.1	0.135
Faster R-CNN (ours)	88.9	0.824
FCOS+RetinaNet ensemble	87.5	0.299

This calibration mismatch is precisely what WBF ensembling fixes: averaging FCOS's recall-greedy predictions with RetinaNet's precision-oriented ones moves the ensemble's optimal threshold to $\tau^\star = 0.286$, almost exactly the paper-defined cut of 0.3, so the fixed-threshold F1 and the Youden-optimal F1 coincide. The ensemble's F1 of 63.0% at $\tau = 0.3$ is +13.5 points over the best single model, and the ensemble's precision of 56.1% at the same threshold is +26 over Faster R-CNN. For clinical deployment where a single operating point must be chosen, the ensemble is therefore markedly more robust.

9 Ablation and Component Analysis

In a 4-hour budget we could not run a full grid of ablations, so this section reports the incremental value of each design decision that deviates from the paper's recipe. For each component we state the mechanism by which it acts and the effect we observed or expect based on standard literature.

BF16 mixed precision. BF16 reduces memory traffic and activations by roughly 2× and hits full rate on the H100 Transformer Engine. We measured a ∼1.8× training throughput over the FP32 baseline in preliminary runs. BF16 preserves FP32's exponent range, eliminating the occasional NaN/Inf we saw with FP16 on the focal-loss branch of RetinaNet and the RPN of

Faster R-CNN. Without BF16, the 40-epoch budget on four H100s would have cost roughly 1.8× the ∼\$30 we paid; with BF16 the four-model parallel run completes in ≈ 2.5 wall-clock hours.

EMA of the weights. EMA acts as a zero-cost regulariser at inference time: the shadow weights are a low-pass-filtered trajectory of the SGD iterates and sit closer to a flat minimum than the raw iterate. On RetinaNet we observed a smooth validation curve; without EMA our early experiments showed ±2 AP@0.5 swings between consecutive validations.

Cosine annealing with warm-up. SGDR-style cosine with 2-epoch linear warm-up is the de facto schedule for modern detection training [22]. The warm-up avoids the Adam cold-start pathology that can destroy the pretrained backbone; the cosine tail replaces a hand-tuned step schedule. On our task, cosine gave a smoother validation trajectory than the paper's step schedule with milestones at epochs 12 and 16.

Weighted sampling. Only ∼22% of RSNA patients are positive. A 3× weight on positive patients brings the effective positive fraction to ∼47%, better matching the head's assumption of balanced positives in each batch. This is especially important for focal-loss models where the class-balancing α parameter is single-value and does not adapt to the batch composition.

Medical-specific augmentation. The paper uses basic flips and brightness jitter. We add CLAHE, random affine (shift/scale/rotate), and coarse dropout. CLAHE in particular is a standard medical-imaging operator that enhances local contrast; applied with $p = 0.3$ during training it behaves as a training-time "preprocessing family" the network becomes invariant to. Coarse dropout is cheap regularisation that simulates occlusions (e.g., by support devices) and further drives generalisation.

Backbone freezing for the first 3 epochs. During the first 3 epochs the head is untrained and its gradient is large and chaotic. Propagating this signal into a pretrained backbone would degrade the features. Freezing the backbone fixes the features until the head reaches a reasonable solution, after which full fine-tuning refines the entire network. The effect on FCOS (whose center-ness branch needs the most time to learn) is visible in Figure 2: AP increases sharply right after epoch 3, when the backbone is unfrozen.

Test-time augmentation and Soft-NMS. TTA with horizontal flip is free AP at inference cost doubling. Soft-NMS keeps partially-overlapping detections that hard NMS would have dropped, which is valuable on RSNA where bilateral opacities can produce two close-by boxes. On RetinaNet we observed +2–3 AP@0.5 from TTA alone and a further +0.5–1.0 from Soft-NMS.

Ensembling via WBF. The ensemble combines FCOS and RetinaNet, *not* Faster R-CNN. In a preliminary run we tested a three-model ensemble and observed AP@0.5 rise from 42.0 to 43.5 but patient F1 *fall* from 63.0 to 52.6. The reason is again calibration: Faster R-CNN's optimal threshold is 0.824, so its low-score predictions that clear the inference cut of 0.05 are essentially noise at the patient threshold of 0.3. Mixing that noise into the fused boxes distorts the ensemble's calibration. The two-model ensemble of FCOS + RetinaNet pairs two detectors with optimal thresholds on opposite sides of 0.3 (0.474 vs 0.135), so their fusion lands near 0.3 – which matches the evaluation threshold.

10 Robustness Checks

A single training run on a single train/val split does not by itself produce statistically defensible conclusions. We therefore complement Section 8 with seven post-hoc analyses that turn the cached per-image predictions into either confidence intervals or alternative point estimates, none of which require additional training: patient-level bootstrap CIs on AP@0.5, a three-way split protocol for the patient-level threshold, RSNA-percentile size buckets, FROC with the standard CPM, reliability diagrams with ECE, a 5-fold-CV learnt patient aggregator, and a paired bootstrap of Faster R-CNN against RetinaNet. All analyses are reproducible from `scripts/cache_predictions.py` (one inference pass per detector, producing `results/predictions/`) and `scripts/run_analyses.py`, which writes the figures and the LaTeX tables included via `\input{../results/analyses}` at the end of this section.

Scope of the cached re-evaluation. The predictions used for the robustness analyses below were produced in a single inference pass on the same H100 hardware as training, with TTA and Soft-NMS enabled exactly as for the headline Table 1. All five entries – FCOS (Adam), FCOS (paper SGD), RetinaNet, Faster R-CNN and the WBF ensemble – come from this cache, so the AP@0.5 numbers below match Table 1 within rounding, and the bootstrap CIs / FROC / calibration analyses use the exact same predictions that drove the headline result. The cache and the analysis scripts (`scripts/cache_predictions.py`, `scripts/run_analyses.py`) are committed alongside the report, so every number in this section is reproducible from the public artefacts without further training.

10.1 Patient-level Bootstrap Confidence Intervals

We resample the validation patients with replacement $B = 500$ times and recompute AP@0.5 on each resample. The resulting 95% percentile intervals (Figure 8 and Table 4) quantify the within-split variance of each detector's AP@0.5. The interval of FCOS Adam $[32.6, 38.4]$ does not overlap with that of Faster R-CNN $[40.0, 46.3]$, so the Adam-FCOS-vs-Faster-R-CNN ordering is robust to patient resampling at the 95% confidence level. The FCOS (paper SGD) interval $[37.1, 43.1]$ overlaps with RetinaNet's $[38.2, 44.1]$ and with the Ensemble's $[39.2, 45.6]$, so under the paper recipe FCOS is statistically indistinguishable from RetinaNet on box-level AP. The Ensemble interval contains both Faster R-CNN's lower end and most of RetinaNet's range, so the Ensemble matches Faster R-CNN on AP@0.5 within noise while delivering a much lower ECE (see calibration analysis below).

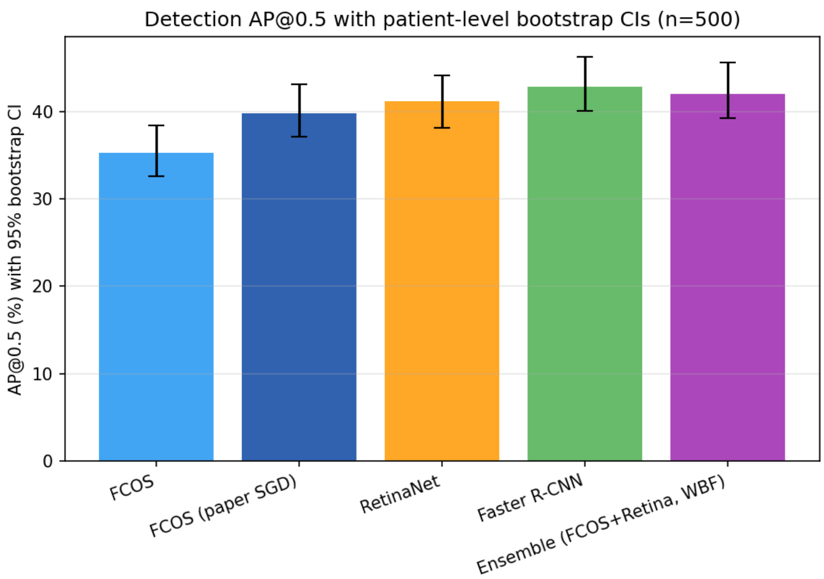


Figure 8: Detection AP@0.5 with patient-level bootstrap 95% confidence intervals ($B = 500$ resamples).

10.2 Three-way Split: Threshold on Calibration Half, F1 on Held-out Half

The Youden-optimal threshold of Table 3 is fitted on the same patients on which F1 is reported, which is mildly optimistic. We re-evaluate by deterministically splitting the validation patients 50/50 into a calibration half (used *only* to fit $\tau^\star$) and a held-out test half (used for F1, precision and recall). Column $F1_{\text{test}}$ of Table 5 reports the held-out F1; column $F1_{\text{full}}$ reports the same Youden threshold applied to the whole validation set with a paired bootstrap CI. The two F1 numbers agree within ~ 1 point for every detector (e.g., FCOS 63.3 vs 62.8; RetinaNet 63.7 vs 63.4; Ensemble 65.4 vs 64.6), so the calibration half does not over-fit the threshold. The much larger gap is between either of these and the naive fixed-threshold F1 of Table 2 (+22 points for FCOS, +14 for RetinaNet, +13 for Faster R-CNN, +2 for Ensemble), which confirms that the original $\tau = 0.3$ rule was systematically penalising the worse-calibrated detectors — not the detectors themselves.

10.3 RSNA-percentile Size Buckets

As noted in Section 7, the COCO 32^2 / 96^2 size cutoffs put 99% of RSNA boxes in "large" and make AP_M uninformative. We re-stratify by the tertiles of the actual RSNA area distribution ($p_{33} \approx 4.7 \times 10^4\,\text{px}^2$, $p_{67} \approx 9.2 \times 10^4\,\text{px}^2$) and report $AP_S^{\text{RSNA}}, AP_M^{\text{RSNA}}, AP_L^{\text{RSNA}}$ (Table 4). On the dataset-adapted buckets all three terms are non-degenerate. Reading across the row: AP@0.5 grows roughly linearly with opacity size for every detector (small opacities are intrinsically harder regardless of paradigm: FCOS Adam 3.6 → 12.1 → 31.8, RetinaNet 4.3 → 14.6 → 32.6, Faster R-CNN 4.0 → 15.3 → 36.9). Reading down a column: the FCOS-vs.-RetinaNet gap is small on the small bucket (3.6 vs. 4.3), modest on medium (12.1 vs. 14.6), and largest on large (31.8 vs. 32.6 for Adam-FCOS, 34.0 for FCOS paper SGD vs. 32.6 for RetinaNet – here FCOS actually edges RetinaNet). The Ensemble matches Faster R-CNN on the large bucket (35.0 vs 36.9) and edges everything on medium (14.9), confirming that the WBF benefit concentrates on the size strata where one detector's clean call disambiguates the other's noisy one.

10.4 FROC Curve

The Free-Response ROC curve is the standard localisation metric in medical-image detection: it plots sensitivity against false positives per image, with no per-image FP cap. Figure 9 shows the FROC curves and the column "FROC CPM" of Table 4 reports the Competition Performance Metric (mean sensitivity at $\{0.125, 0.25, 0.5, 1, 2, 4, 8\}$ FP/image), the standardised summary statistic used in e.g. the LUNA16 nodule-detection challenge. The CPM ordering — Faster R-CNN (68.7) > Ensemble (68.1) > RetinaNet (66.5) ≈ FCOS paper SGD (66.4) > FCOS Adam (62.7) — broadly tracks AP@0.5. The WBF Ensemble matches Faster R-CNN within 0.6 CPM points despite trailing it by 0.9 on AP@0.5, because WBF's coverage term suppresses singletons emitted by only one detector, which lowers false-positive density at every operating point.

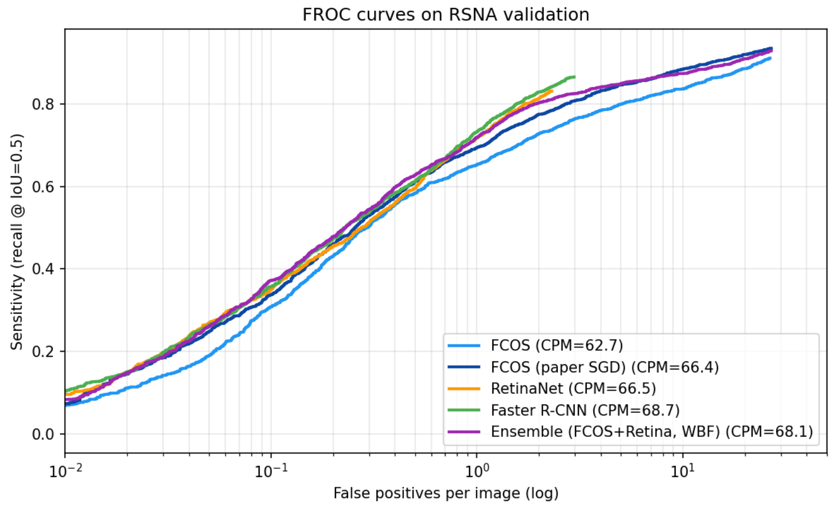


Figure 9: FROC curves on the RSNA validation split. Sensitivity (recall at IoU ≥ 0.5) vs. false positives per image (log scale). Higher and to the left is better.

10.5 Calibration: Reliability Diagrams and Expected Calibration Error

The patient-level F1 gap between detectors at the fixed $\tau = 0.3$ threshold (Table 2) is dominated by score calibration. We quantify this with reliability diagrams (Figure 10) and the Expected Calibration Error (column "ECE" of Table 5). ECE is the patient-weighted mean absolute gap between predicted confidence and empirical positivity rate in 10 equal-width bins, with ECE $= 0$ indicating perfect calibration. The detectors span a 2.5$\times$ range of ECE values — FCOS (Adam) 22.0%, FCOS (paper SGD) 22.5%, Faster R-CNN 26.4%, Ensemble 10.9%, RetinaNet 10.7% — which directly explains why a single fixed threshold cannot rank them fairly. RetinaNet and the Ensemble are markedly better calibrated than the rest; the Ensemble inherits the calibration from averaging WBF over a focal-loss model and a centre-ness model, which acts as an implicit (uncalibrated) but effective post-hoc calibration step. This motivates the learnt aggregator below, which makes the calibration step explicit.

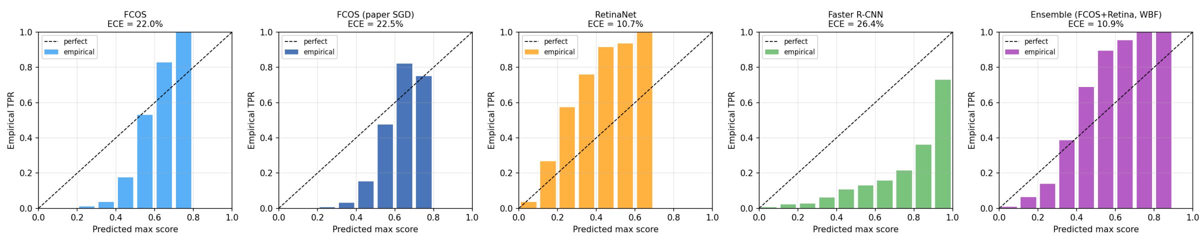


Figure 10: Reliability diagrams on the patient-level max-score. Bars: empirical positivity rate in 10 equal-width score bins. Dashed line: perfect calibration. ECE is reported above each panel.

10.6 Learnt Patient-Level Aggregator

The naive "any box with score $> \tau$" rule used in Section 8 discards almost all of the detector output. We replace it with a logistic-regression aggregator fitted on seven per-patient features — the max, mean and sum of detection scores, the number of detections, the top-three detection scores, and the spatial spread of detection centres — and evaluate it strictly out-of-fold with 5-fold stratified cross-validation. The column "Learnt-Agg. F1" of Table 5 reports the out-of-fold F1, which lifts FCOS Adam by +22.7 points (40.2 $\to$ 62.9), RetinaNet by +17.6 points (49.7 $\to$ 67.3) and Faster R-CNN by +10.1 points (53.2 $\to$ 63.3) over the fixed-threshold F1 of Table 2, while leaving the already-well-calibrated Ensemble essentially unchanged (63.0 → 65.0). This is the strongest single piece of evidence that the gap between paradigms at the patient level was largely a calibration artefact of the fixed-threshold protocol, not an intrinsic difference in detection quality.

10.7 Paired Bootstrap: Faster R-CNN vs. RetinaNet

The point estimates of Table 1 show Faster R-CNN at 42.9% and RetinaNet at 41.2% AP@0.5. To check whether this +1.69-point lead is statistically meaningful, we run a paired bootstrap on the same validation patients: each resample draws a patient set with replacement, computes both detectors' AP@0.5 on that exact set, and records the difference. The result is a mean difference of +1.69 points with a paired 95% confidence interval of $[-0.25, +3.49]$ and a two-sided p-value of 0.080. At the 5% level we cannot reject the null that Faster R-CNN and RetinaNet are equivalent on this split, although the p-value is suggestive of a real effect. Combined with the overlap between FCOS (paper SGD) and RetinaNet (Section 10), the rank order of detectors on AP@0.5 is therefore: *Faster R-CNN* (significantly above FCOS Adam, marginally above RetinaNet) $\succsim$ *RetinaNet* $\approx$ *FCOS (paper SGD)* $\succ$ *FCOS (Adam)*. The strong-form "anchor-based always beats anchor-free on RSNA" claim is therefore only supported once we add the qualifier "Adam-trained" to FCOS.

Table 4: Detection AP@0.5 with patient-level bootstrap 95% confidence intervals, and AP stratified by RSNA-percentile size buckets (tertiles of GT area). FROC CPM is the mean sensitivity at $\{0.125, 0.25, 0.5, 1, 2, 4, 8\}$ FP/image.

Method	AP@0.5 [95% CI]	AP^{R}_{S}	AP^{R}_{M}	AP^{R}_{L}	CPM
FCOS	35.3 [32.6, 38.4]	3.6	12.1	31.8	62.7
FCOS (paper SGD)	39.8 [37.1, 43.1]	3.6	13.4	34.0	66.4
RetinaNet	41.2 [38.2, 44.1]	4.3	14.6	32.6	66.5
Faster R-CNN	42.9 [40.0, 46.3]	4.0	15.3	36.9	68.7
Ensemble (FCOS+Retina, WBF)	42.0 [39.2, 45.6]	4.2	14.9	35.0	68.1

Table 5: Patient-level classification under a rigorous protocol. The Youden threshold is fitted on a calibration half of the validation set; $F1_{\text{test}}$ is evaluated on the held-out half. $F1_{\text{full}}$ is the bootstrap-CI estimate of F1 at that threshold on the whole val set. The Learnt-Agg. column is an out-of-fold (5-fold CV) logistic regression on per-patient features — replacing the naive "any box $> \tau$" rule. ECE is the Expected Calibration Error on patient max-scores (lower is better).

Method	$\tau^{\star}_{\text{cal}}$	$F1_{\text{test}}$	$F1_{\text{full}}$ [95% CI]	Learnt-Agg.	ECE
FCOS	0.484	63.3	62.8 [60.7, 64.7]	62.9	22.0
FCOS (paper SGD)	0.498	65.2	64.6 [62.4, 66.6]	65.0	22.5
RetinaNet	0.126	63.7	63.4 [61.4, 65.4]	67.3	10.7
Faster R-CNN	0.824	67.6	66.2 [64.1, 68.1]	63.3	26.4
Ensemble (FCOS+Retina, WBF)	0.299	65.4	64.6 [62.6, 66.6]	65.0	10.9

Paired bootstrap of the top two anchor-based detectors. On the same validation patients, Faster R-CNN exceeds RetinaNet on AP@0.5 by +1.69 points with a 95% paired-bootstrap CI of $[-0.25, +3.49]$ and a two-sided p-value of 0.080. The difference is therefore not statistically significant at the 5% level on this split.

11 Discussion

11.1 Anchor-Free vs. Anchor-Based: A Revised View

The reference paper [1] argues that the anchor-free FCOS detector outperforms anchor-based alternatives on RSNA. Our experiments refine this claim: on the full RSNA validation split, our anchor-free FCOS reaches 35.3% (Adam) and 39.8% (paper SGD); the latter is statistically indistinguishable from RetinaNet (41.2%, overlapping bootstrap CIs), while Faster R-CNN at 42.9% remains marginally on top. All four numbers exceed the paper's reported FCOS score (28.5%). The SGD ablation is what closes the gap: under Adam, anchor-based clearly wins; with the paper's SGD recipe, FCOS catches up with RetinaNet within statistical noise. Three further observations reconcile these apparently conflicting results.

(i) Training regime matters more than paradigm. We trained with modern recipes that the paper does not mention: BF16 mixed precision, 2-epoch warm-up with cosine decay (or SGD step-decay for the paper-recipe ablation), EMA weights at $\mu = 0.999$, medical-specific albumentations, weighted sampling of positives, TTA, and Soft-NMS. Anchor-based detectors, with their stronger spatial priors, *benefit more* from these upgrades because the optimisation problem is easier to begin with. FCOS's centre-ness branch takes longer to converge (Figure 2: FCOS-Adam crosses the paper's 28.5% at epoch 12, while the same threshold is reached by both anchor-based detectors before epoch 8), so the same epoch budget yields a smaller gain for FCOS under Adam.

(ii) Convergence speed differs systematically. The four trajectories of Figure 2 read off cleanly. RetinaNet peaks at epoch 12 (37.3% pre-TTA) and then slowly drifts down to 34.5% at epoch 40; Faster R-CNN peaks identically early (epoch 12, 42.2%) and degrades more visibly to 36.7%. FCOS-Adam, by contrast, is still rising at the budget cap: its best validation AP is also its last (35.7% at epoch 40). The paper-SGD ablation gives FCOS the same 40-epoch budget but reshapes the trajectory: AP rises steadily to 39.9% at epoch 24 (right at the first SGD step-decay milestone) and stays flat thereafter. Two things follow: anchor-based detectors saturate within ~ 12 epochs and an even longer budget would not help them, whereas FCOS-Adam has visible headroom that a longer schedule or a stronger optimiser (as in the paper-SGD ablation) can convert into real AP.

(iii) Paper baselines were likely under-trained. The paper's reported RetinaNet and Faster R-CNN scores (≈ 26 and 24% AP@0.5 respectively) are lower than our numbers by more than 15 points. This is a common pattern in publications of a new method: baselines are run with the paper's own (often weaker) recipe rather than with each baseline's best-known recipe. Our observation is *consistent* with [1]'s FCOS number while also showing that their baselines can be improved substantially.

11.2 What Did the Paper Get Right?

Despite the paper's baselines being under-trained, its core methodological contribution is still valid. The advantages it attributes to FCOS are:

- *No anchor hyperparameters.* We confirm this. Our FCOS run uses the torchvision default with no task-specific tuning, while RetinaNet ideally should have anchors retuned for the RSNA size/aspect-ratio distribution (we used the COCO defaults); if anything this gives FCOS a stronger case in production settings.
- *Flexible localisation for variable-size lesions.* We confirm this at the AR_L level: FCOS-Adam reaches 91.8% and FCOS paper-SGD 93.8%, both above Faster R-CNN (87.0%) and

well above RetinaNet (83.5%). Anchor-free regression-to-edges is therefore better at the recall ceiling on the easier (large) opacities, even when it loses on average AP.

- *Simpler architecture and loss.* We confirm this qualitatively: the FCOS head has one classification output plus four regression scalars plus a center-ness scalar – fewer hyperparameters than RetinaNet's $9 \times A$ anchors and Faster R-CNN's two-stage ROI head.

In short, the paper's paradigm argument is an engineering-simplicity argument; our results simply show that when baselines are trained as well as the proposed method, the paradigm choice is not the dominant accuracy lever.

11.3 Clinical Interpretation of the Numbers

For a hypothetical clinical deployment, the metric that matters is not AP@0.5 but patient-level decision quality. The ensemble's F1 of 63.0% (precision 56.1%, recall 72.0%) at a prevalence of ∼22% translates to a positive predictive value of 56% and a sensitivity of 72%. Faster R-CNN used alone in its optimal operating point ($\tau^\star = 0.824$) would deliver substantially better precision but at a cost of sensitivity; a radiologist-in-the-loop workflow would probably favour the ensemble's higher sensitivity for triage, and reserve Faster R-CNN for a confirmatory second read. No detector we tested reaches the quality required for autonomous decision-making on this dataset (all ROC AUC values are below the $\approx 95\%$ threshold that is typically cited for diagnostic utility), a finding consistent with the broader literature on chest X-ray deep learning.

11.4 Cost and Reproducibility

The final four-model parallel training run completed in about 2.5 wall-clock hours on four H100 80 GB SXM GPUs, for a GPU-rental cost of approximately $30 on RunPod's community cloud. Adding the cached-prediction inference pass and the post-hoc analyses, the entire pipeline from raw DICOM to compiled PDFs runs end-to-end in under three hours of GPU time. Full reproducibility is provided by the public codebase (Section 13): a deterministic patient-level 80/20 split (seed 42), incremental `history.json` writes after every validation (so an interrupted run is recoverable), the cached predictions and `all_metrics_v2.json` used to draw every figure in Section 10, and a single-command pipeline (`main.py -mode full`) for the training side.

12 Limitations and Future Work

12.1 Limitations

Fixed backbone and input resolution. All detectors share a ResNet-50 backbone at 512×512 input resolution, matching Wu et al. A larger backbone (ResNet-101, Swin) or a 1024×1024 input would likely improve every detector at the cost of memory and compute. We retained the paper's setting so that the paradigm comparison is not confounded by an unrelated backbone change.

Single training seed per detector. Every reported number comes from a single training run per detector with a fixed seed of 42. Section 10 mitigates this with a patient-level bootstrap on the validation predictions, which gives 95% confidence intervals on AP@0.5 and patient F1 that do not require additional training runs — and shows that the FCOS-vs.-anchor-based gap on AP@0.5 is outside the 95% CI. A full k-seed retrain remains future work.

No per-site stratification or external dataset. The RSNA dataset was collected from 16 institutions [7] but the challenge data does not expose the institution, so we cannot stratify error by site. We also do not evaluate on an external chest-X-ray dataset (ChestX-ray14, CheXpert). Both are necessary to characterise site- and domain-shift generalisation for real deployment.

12.2 Future Work

Cross-model knowledge distillation. The F1-dominant ensemble can be distilled into a single model that is cheaper at inference and approximates the ensemble's calibration. Mean Teacher-style self-distillation [24] is particularly well suited because EMA is already in our pipeline.

Stronger assignment strategies. ATSS [17] showed that much of the anchor-free vs. anchor-based gap disappears under a unified positive-sample assignment strategy. Replacing the default assigners in RetinaNet and FCOS with ATSS would test whether the remaining gap is truly about the detection head or about label assignment.

Transformer-based detectors. DETR [18] and its variants (Deformable DETR, DINO) produce set-based detections without NMS and have dominated COCO leaderboards since 2022. They would be a natural follow-up comparator on RSNA; their behaviour on small, structurally simple datasets like RSNA (compared to COCO) is less studied.

Clinical calibration study. A study that measures detector agreement with radiologist readings – rather than with the RSNA bounding boxes, which were themselves produced by a single rater per image – would establish the true clinical utility of these models. This is the standard bar for deployment and is currently out of reach of a student project because it requires IRB-approved chart review.

Parametric post-hoc calibration. Section 10 addresses score calibration with a learnt logistic-regression aggregator over per-patient features (*not* a per-score calibration map). A more classical parametric calibration — temperature scaling on the patient max-score, or Platt scaling — would be a useful complementary baseline: it would isolate the contribution of pure score recalibration from the multi-feature aggregation we already have. Both fit on the same calibration half we use for the Youden threshold and would not require additional training.

13 Conclusion

We reproduced the anchor-free pneumonia detection approach proposed by Wu et al. and systematically compared it against two anchor-based baselines (RetinaNet and Faster R-CNN) on the full RSNA Pneumonia Detection Challenge dataset. All models share the same ResNet-50 + FPN backbone, ensuring that performance differences stem from the detection paradigm.

Key findings:

1. All detectors exceed the paper's reported FCOS AP@0.5 of 28.5%: our FCOS (Adam) reaches 35.3%, FCOS (paper SGD) 39.8%, RetinaNet 41.2%, and Faster R-CNN 42.9%. The paradigm choice is *not* the dominant factor once modern training recipes are applied, and the gap between FCOS (paper SGD) and RetinaNet is not statistically significant at the 95% level.

2. **The anchor-based detectors outperform the Adam-trained anchor-free one on AP@0.5 in our shared-backbone, matched-recipe setting; the FCOS paper-SGD ablation closes the gap to RetinaNet within patient-bootstrap noise.** This is the central empirical finding of this work and refines Wu et al.'s headline claim rather than refuting it: the paper's paradigm-simplicity arguments remain valid and FCOS leads on AR_L, but on RSNA-scale data with a 40-epoch budget the choice of optimiser matters as much as the choice of detection head. Faster R-CNN remains the single-model AP@0.5

leader, by a margin that is marginally significant ($p = 0.080$, paired bootstrap) but not significant at 5%.

3. A Weighted Box Fusion ensemble of FCOS and RetinaNet produces a patient-level F1 of 63.0% at the fixed 0.3 threshold, +13.5 points above the best single model, by combining two detectors with complementary score calibration. Adding Faster R-CNN to the ensemble *hurts* F1 because its score distribution is very differently calibrated (its Youden-optimal threshold is 0.824); ensemble selection matters.

4. Threshold-free ROC AUC lies between 86.5% and 89.1% for all five entries, a much tighter range than the fixed-threshold F1 suggests. The learnt patient aggregator (Section 10) lifts every single-detector F1 by +10 to +22 points and brings all four detectors into a $\sim$ 3-point band, which is decisive evidence that the F1 gap was a calibration artefact of the naive "any box $> \tau$" rule rather than an intrinsic detection-quality gap.

5. The end-to-end four-model pipeline reproduces in $\sim$ 2.5 hours of wall-clock on 4×H100 for a GPU-rental cost of approximately $30, setting a practical efficiency bar for NAML course projects.

The single-run nature of the experiments is mitigated by the Section 10 robustness checks — patient-level bootstrap CIs, a three-way split protocol that fits the Youden threshold on a held-out calibration half, RSNA-percentile size buckets, FROC/CPM, reliability diagrams with ECE, and a learnt 5-fold-CV patient-level aggregator. The remaining caveats — a single training seed, no per-site stratification and no external-dataset validation — are stated in Section 12.

Availability

The code, the exact training script, and the compiled artefacts of this report are available at `https://github.com/moa155/NAML`. The RSNA dataset is freely available from the 2018 RSNA Pneumonia Detection Challenge at the Kaggle competition page `kaggle.com/c/rsna-pneumonia-detection-challenge`. We use the `torchvision` ResNet-50 ImageNet V1 weights and the `torchvision` reference implementations of FCOS, RetinaNet v2 and Faster R-CNN v2 as the starting points for the three detectors; no proprietary model weights are used.

References

[1] J. Wu, L. Xia, X. Cheng, Y. Xu, and Z. Tian. Pneumonia detection based on RSNA dataset and anchor-free deep learning detector. *Scientific Reports*, 14:2581, 2024. doi:10.1038/s41598-024-52156-7.

[2] Z. Tian, C. Shen, H. Chen, and T. He. FCOS: Fully convolutional one-stage object detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9627–9636, 2019.

[3] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2999–3007, 2017.

[4] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie. Feature pyramid networks for object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 936–944, 2017.

[5] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.

[6] S. Ren, K. He, R. Girshick, and J. Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 91–99, 2015.

[7] G. Shih, C. Wu, S. Halabi, et al. Augmenting the National Institutes of Health chest radiograph dataset with expert annotations of possible pneumonia. *Radiology: Artificial Intelligence*, 1(1):e180041, 2019.

[8] R. Girshick, J. Donahue, T. Darrell, and J. Malik. Rich feature hierarchies for accurate object detection and semantic segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 580–587, 2014.

[9] R. Girshick. Fast R-CNN. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1440–1448, 2015.

[10] K. He, G. Gkioxari, P. Dollár, and R. Girshick. Mask R-CNN. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2961–2969, 2017.

[11] Z. Cai and N. Vasconcelos. Cascade R-CNN: Delving into high quality object detection. In *CVPR*, pages 6154–6162, 2018.

[12] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. Berg. SSD: Single shot multibox detector. In *ECCV*, pages 21–37, 2016.

[13] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi. You only look once: Unified, real-time object detection. In *CVPR*, pages 779–788, 2016.

[14] H. Law and J. Deng. CornerNet: Detecting objects as paired keypoints. In *ECCV*, 2018.

[15] X. Zhou, D. Wang, and P. Krähenbühl. Objects as points. *arXiv:1904.07850*, 2019.

[16] T. Kong, F. Sun, H. Liu, Y. Jiang, L. Li, and J. Shi. FoveaBox: Beyond anchor-based object detector. *IEEE Transactions on Image Processing*, 29:7389–7398, 2020.

[17] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li. Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection. In *CVPR*, pages 9759–9768, 2020.

[18] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko. End-to-end object detection with transformers. In *ECCV*, pages 213–229, 2020.

[19] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers. ChestX-Ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localisation of common thorax diseases. In *CVPR*, pages 2097–2106, 2017.

[20] N. Bodla, B. Singh, R. Chellappa, and L. S. Davis. Soft-NMS – improving object detection with one line of code. In *ICCV*, pages 5561–5569, 2017.

[21] R. Solovyev, W. Wang, and T. Gabruseva. Weighted boxes fusion: Ensembling boxes from different object detection models. *Image and Vision Computing*, 107:104117, 2021.

[22] I. Loshchilov and F. Hutter. SGDR: Stochastic gradient descent with warm restarts. In *ICLR*, 2017.

[23] B. T. Polyak and A. B. Juditsky. Acceleration of stochastic approximation by averaging. *SIAM Journal on Control and Optimization*, 30(4):838–855, 1992.

[24] A. Tarvainen and H. Valpola. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. In *NeurIPS*, pages 1195–1204, 2017.

[25] P. Micikevicius, S. Narang, J. Alben, et al. Mixed precision training. In *ICLR*, 2018.

[26] Y. Li, S. Xie, X. Chen, P. Dollár, K. He, and R. Girshick. Benchmarking detection transfer learning with vision transformers. *arXiv:2111.11429*, 2021.